Module 3

1. Different 802.11 PHY Layer Standards and Their Characteristics

The PHY layer defines how Wi-Fi signals are transmitted over radio waves.
Different 802.11 standards improve speed, range, frequency usage, and efficiency.

Standard	Frequency	Max Speed	Technology
802.11a	5 GHz	54 Mbps	OFDM
802.11b	2.4 GHz	11 Mbps	DSSS
802.11g	2.4 GHz	54 Mbps	OFDM
802.11n	2.4/5 GHz	600 Mbps	MIMO + OFDM
802.11ac	5 GHz	Several Gbps	MU-MIMO
802.11ax	2.4/5/6 GHz	Higher efficiency	OFDMA + MU-MIMO
802.11be	2.4/5/6 GHz	Extremely high	Multi-Link Operation

Newer standards mainly improve:

- Speed
- Capacity
- Multiple user handling
- Reduced latency

2. DSSS and FHSS

DSSS (Direct Sequence Spread Spectrum)

DSSS spreads data over a wider frequency range using a spreading code.
Each bit is converted into multiple smaller bits called chips.
This improves resistance to interference and noise.

Used in:

- 802.11b

Advantage:

- Better reliability

Disadvantage:

- Lower speed

FHSS (Frequency Hopping Spread Spectrum)

FHSS rapidly changes transmission frequency according to a hopping pattern.
Both sender and receiver hop together synchronously.

Advantages:

- Resistant to interference
- More secure

Disadvantages:

- Lower throughput
- Less efficient for modern WLANs

Used in:

- Early Wi-Fi systems

3. Modulation Schemes in PHY Layer

Modulation converts digital bits into radio signals.

Common modulation schemes:

Modulation	Bits per Symbol	Performance

BPSK	1	Very reliable, slow
QPSK	2	Moderate speed
16-QAM	4	Faster
64-QAM	6	High speed
256-QAM	8	Very high speed
1024-QAM	10	Used in Wi-Fi 6

Higher modulation:

- Gives higher speed
- Requires cleaner signal/SNR

Lower modulation:

- More reliable
- Works at longer range

4. Significance of OFDM in WLAN

OFDM stands for Orthogonal Frequency Division Multiplexing.

Instead of using one large channel, OFDM divides the channel into many smaller subcarriers.

Benefits:

- Better spectral efficiency
- Less interference
- Better multipath handling
- Higher throughput

Used in:

- 802.11a/g/n/ac/ax

OFDM greatly improved Wi-Fi performance in indoor environments.

5. Wi-Fi Frequency Bands and Channels

Wi-Fi operates mainly in:

Band	Features
2.4 GHz	Long range, more interference
5 GHz	Higher speed, less interference
6 GHz	Wi-Fi 6E/7, very clean spectrum

2.4 GHz Channels

Channels overlap heavily.
Non-overlapping channels:

- 1
- 6
- 11

5 GHz

Many more channels available:

- Less congestion
- Supports wider channels

6 GHz

Newest band:

- Large bandwidth
- Low latency
- Better for high-density deployments

6. Guard Interval in WLAN

Guard Interval (GI) is a small time gap inserted between OFDM symbols.

Purpose:

- Prevent overlap between symbols
- Reduce multipath interference

Normal GI:

- 800 ns

Short GI:

- 400 ns

Short GI improves efficiency because less time is wasted between symbols, increasing data rate slightly.

7. Structure of 802.11 PHY Layer Frame

The PHY frame is called **PPDU (PLCP Protocol Data Unit)**.

Main parts:

1. **Preamble**
    a. Helps synchronization
    b. Signal detection
2. **PHY Header**
    a. Contains rate and length info
3. **Payload/Data**
    a. Actual MAC frame

The receiver uses the preamble to properly decode the signal.

8. OFDM vs OFDMA

OFDM	OFDMA
One user uses entire channel	Multiple users share channel
Sequential transmission	Simultaneous transmission
Used in older Wi-Fi	Used in Wi-Fi 6
Less efficient in crowded networks	Better efficiency

OFDMA divides subcarriers into Resource Units (RUs) for multiple users.

9. MIMO vs MU-MIMO

MIMO

Multiple antennas communicate with one client at a time.

Purpose:

- Increase speed
- Improve reliability

MU-MIMO

Multiple users communicate simultaneously.

Advantages:

- Better efficiency
- More users handled together

Used heavily in:

- Wi-Fi 5 and Wi-Fi 6

10. PPDU, PLCP, and PMD

PPDU

Physical frame transmitted over air.

Contains:

- Preamble
- PHY header
- Data

PLCP (Physical Layer Convergence Procedure)

Acts as interface between:

- MAC layer
- Physical transmission layer

Responsibilities:

- Frame formatting
- Header creation

PMD (Physical Medium Dependent)

Handles actual radio transmission.

Functions:

- Modulation
- Coding

- Signal transmission
- Frequency operations

11. Types of PPDU Across Wi-Fi Generations

Different Wi-Fi generations use different PPDU formats.

Wi-Fi Version	PPDU Type
802.11a/g	Legacy PPDU
802.11n	HT PPDU
802.11ac	VHT PPDU
802.11ax	HE PPDU

Each newer PPDU supports:

- Higher efficiency
- More antennas
- Better modulation
- OFDMA/MU-MIMO features

12. How Data Rate is Calculated

Wi-Fi data rate depends on:

- Channel width
- Modulation
- Coding rate
- Number of spatial streams
- Guard interval

Basic idea:

$\text{Data Rate} \propto \text{Channel Width} \times \text{Bits per Symbol} \times \text{Spatial Streams}$ *\text{Data Rate} \propto \text{Channel Width} \times \text{Bits per Symbol} \times*

\text{Spatial Streams}$\text{Data Rate} \propto \text{Channel Width} \times \text{Bits per Symbol} \times \text{Spatial Streams}$

Higher:

- Channel width
- QAM level
- Spatial streams

→ gives higher throughput.

Example:

- 20 MHz + 1 stream = lower speed
- 80 MHz + 4 streams + 1024-QAM = very high speed